

ARHAAM HOSSAIN

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Education

Stony Brook University

Bachelor of Engineering in Electrical Engineering

August 2022 – Expected December 2026

Stony Brook, NY

Relevant Coursework

- Digital Logic Design
- Embedded Systems
- Circuit Board Design
- Semiconductor Devices
- Integrated Electronics
- Control Systems Design
- Power Electronics
- VLSI System Design

Technical Skills

Languages: C, C++, MATLAB, Python, AVR Assembly, SystemVerilog, VBA

Developer Tools: Altium Designer, KiCAD, HSPICE, LTspice, OrCAD, PSpice, Virtuoso, Xilinx Vivado

Lab Tools: Oscilloscope, Digital Multimeter, Bench Power Supply, Function Generator, Logic Analyzer, Spectrum Analyzer

Experience

Archer

Incoming Power Engineer Intern

June – August 2026

San Jose, California

Spellman High Voltage

Electrical Engineering Intern

June – August 2025

Hauppauge, New York

- Engineered and validated a custom gate-drive transformer for a SiC FET-based high-voltage solid-state relay (SSR) operating at switching frequencies up to 20 kHz and rated for isolation voltages up to 100 kV.
- Modeled transformer magnetics and SSR switching behavior in LTspice, including leakage inductance and core saturation limits, to predict switching losses and ensure reliable gate-drive operation prior to hardware testing.
- Improved high-voltage reliability by refining gate-drive topology to reduce ringing and overshoot, verified through oscilloscope measurements of gate voltage, drain current, and switching transients.

GE Vernova

Systems Engineering Intern

May – August 2024

Schenectady, New York

- Developed a VBA-based modeling tool to evaluate round-trip efficiency, guaranteed availability, and usable energy capacity for 500+ MW Battery Energy Storage Systems (BESS) by parameterizing different battery characteristics.
- Created an OCV and DC internal resistance model for lithium-ion cells operating in constant power mode, enabling thermal and efficiency analysis under realistic load conditions.
- Built a data processing workflow to analyze vendor cell test data, extract charge/discharge metrics, and benchmark against degradation models, improving life-cycle prediction accuracy.

Projects

Low Power IC for Neuromorphic Computing | Cadence Virtuoso

August – Present

- Designing a low power ReRAM-based compute-in-memory architecture for SNN-based word detection in noisy environments, mapping quantized synaptic weights onto crossbar arrays to enable matrix-vector multiplication (MVM).
- Implementing Transposable Neurosynaptic Array (TNSA) cores using the SkyWater 130 nm PDK, integrating word-line, bit-line, and source-line drivers for forward MVM operation.
- Constructing transistor-level 1T1R ReRAM cells and CMOS neuron/ADC circuits in Cadence Virtuoso, completing schematic capture, layout, and full DRC/LVS verification for tapeout.

FPGA-Accelerated Reactive Collision UAV | MATLAB, SystemVerilog

September 2023 – Present

- Developing a reactive collision UAV using a Zybo Z7: Zynq-7000 FPGA SoC and SEN0634 global-shutter optical-flow camera, offloading frame-based optical flow computation to FPGA, reducing vision-processing latency.
- Implementing a frame-based optical flow pipeline using MATLAB Vision HDL Toolbox, translating algorithmic models into SystemVerilog processing blocks and integrating FPGA motion vectors with control logic for real-time response.
- Validating real-time collision response by testing optical flow outputs and control behavior under dynamic motion, and different lighting, ensuring stable operation without reliance on GPS.

SBRT MATE ROV Motherboard | Altium Designer

January – April 2026

- Designed a 6-layer ROV motherboard integrating power distribution, control, and sensing into a single assembly, supporting ESCs, buck converters, an NVIDIA Jetson, motor drivers, and onboard sensor interfaces.
- Developed an STM32-interfaced sensor suite integrating dual IMUs, a depth sensor, load 12 V/5 V rail monitoring, and leak sensing over I2C, enabling sensor fusion.
- Executed PCB layout with an optimized stack-up ensuring high-current power distribution and signal integrity, implementing large copper pours, stitching vias, and low-impedance return paths.